SECTIONS OF A SHEAF ON A BASIS ARE EXACTLY THE POINTWISE FAMILIES SATISFYING THE LOCAL CONDITION

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Allowed standard prerequisites (editorial).

- Sheaf on a basis: the sheaf condition and its uniqueness part stated for coverings and pairwise intersections by basis elements (definition-sheaf-basis). invoked at proof lines 4127-4129 and 4136-4138. Provides both the injectivity of F(U) into the product of stalks and the gluing of the sigma_i.

Source reference. Original definition 009J: sources/stacks-sheaves/pinned-original-source.tex, source lines 3951–3951. Cited at proof lines 4127-4129 (uniqueness part, injectivity) and again at proof lines 4136-4138 (gluing condition (**)); both uses load-bearing inside the counted span.

Lemma 0.2. *Let X be a topological space. Let $\mathcal{B}$ be a basis for the topology on X. Let $U \in \mathcal{B}$. Let $\mathcal{F}$ be a sheaf of sets on $\mathcal{B}$. The map*

$$\mathcal{F}(U) \to \prod\nolimits_{x \in U} \mathcal{F}_x$$

identifies $\mathcal{F}(U)$ with the elements $(s_x)_{x \in U}$ with the property

(*) *For any $x \in U$ there exists a $V \in \mathcal{B}$, with $x \in V \subset U$ and a section $\sigma \in \mathcal{F}(V)$ such that for all $y \in V$ we have $s_y = (V, \sigma)$ in $\mathcal{F}_y$.*

Proof. First note that the map $\mathcal{F}(U) \to \prod_{x \in U} \mathcal{F}_x$ is injective by the uniqueness in the sheaf condition of Definition 009J. Let (s_x) be any element on the right hand side which satisfies $(*)$. Clearly this means we can find a covering $U = \bigcup U_i$, $U_i \in \mathcal{B}$ such that $(s_x)_{x \in U_i}$ comes from certain $\sigma_i \in \mathcal{F}(U_i)$. For every $y \in U_i \cap U_j$ the sections σ_i and σ_j agree in the stalk $\mathcal{F}_y$. Hence there exists an element $V_{ijy} \in \mathcal{B}$, $y \in V_{ijy}$ such that $\sigma_i|_{V_{ijy}} = \sigma_j|_{V_{ijy}}$. Thus the sheaf condition $(**)$ of Definition 009J applies to the system of σ_i and we obtain a section $s \in \mathcal{F}(U)$ with the desired property. □

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